The Polyhedral Stretching Theorem

Stretch any solid with equal edges one edge-length in any direction and every edge stays the same length: the Platonic solids, DICTO's stretched dodecahedra, three and four caps, and paper models to build them

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Version 1 · 22 October 2026 · DICTO DOI: reserved on publication · Licence CC BY-SA 4.0 · Scripts, checks, figures and printable nets (Swept Dodecahedra): https://github.com/DICTOR-Master/kaleidohedra/tree/4982c91/studies/2026-10-22-swept-dodecahedra (research branch, commit 4982c91; nets and study attached to this record)
The stretched dodecahedron (F11), the long hexagon seen in the dodecahedron layer, the three- and four-cap forms and the telescoping model are DICTO's ideas; computation, proofs and drafting with Claude (Anthropic).

In short. Take a regular dodecahedron and slide it one edge-length in a straight line. The shape it sweeps through has every edge the same length, whichever way you slide it, except when the slide runs exactly along one of its edges. Its faces are the dodecahedron's twelve pentagons and a belt of rhombi, ten or twelve of them; where the slide lies flat along a pentagon, that pentagon stretches into a hexagon or a heptagon. Sliding along a cube axis gives DICTO's stretched dodecahedron (F11); along a five-fold axis the ten rhombi are all alike; sliding along a face diagonal of Euclid's cube gives the member DICTO found, all twelve pentagons and ten rhombi, which began as a void and two overlapping dodecahedra that DICTO saw in the dodecahedron layer of paper #47. Placing dodecahedra at the corners of a triangle or a tetrahedron gives the three- and four-cap forms, with equilateral triangles among their faces. The same theorem holds for every solid whose edges are all equal, so for every Platonic solid; the stretched icosahedra form a second new-looking family, and the tetrahedron and octahedron rediscover the cuboctahedron, the truncated tetrahedron and three Johnson solids. DICTO names the family the **DICTO stretched solids**: **equal-edged** when the stretch is one edge-length, **telescoped** at any other length. Printable nets, and a telescoping model that slides from equal-edged to long, are attached.



Figure 1. The dodecahedron swept along the face diagonal of Euclid's cube to the next dodecahedron: twelve pentagons (violet) and a belt of ten long faces (lilac). Left: along the long edge, the long hexagon DICTO saw. Middle: end on, a decagon. Right: oblique.

1. Where it started

> *DICTO looked at a block of dodecahedra and saw a void, a long six-sided space, with two dodecahedra overlapping it on either side. Taken together, the void and the two overlaps became a stretched dodecahedron.*

In the dodecahedron layer of #47 [1], dodecahedra sit on every other cube of a cubic lattice. Looking into the block, DICTO saw a void, a long hexagon in outline, with two dodecahedra overlapping it on either side, and proposed a dodecahedron variant to fit it. The void and the two overlaps, taken together, make one solid: the space a dodecahedron passes through when it slides from one of the two to the other, along the face diagonal of the cube it is built on (Euclid's cube [2]). DICTO had found a first member of this family earlier, also from overlapped dodecahedra: the stretched dodecahedron, the same slide along a cube axis (F11 [3]).

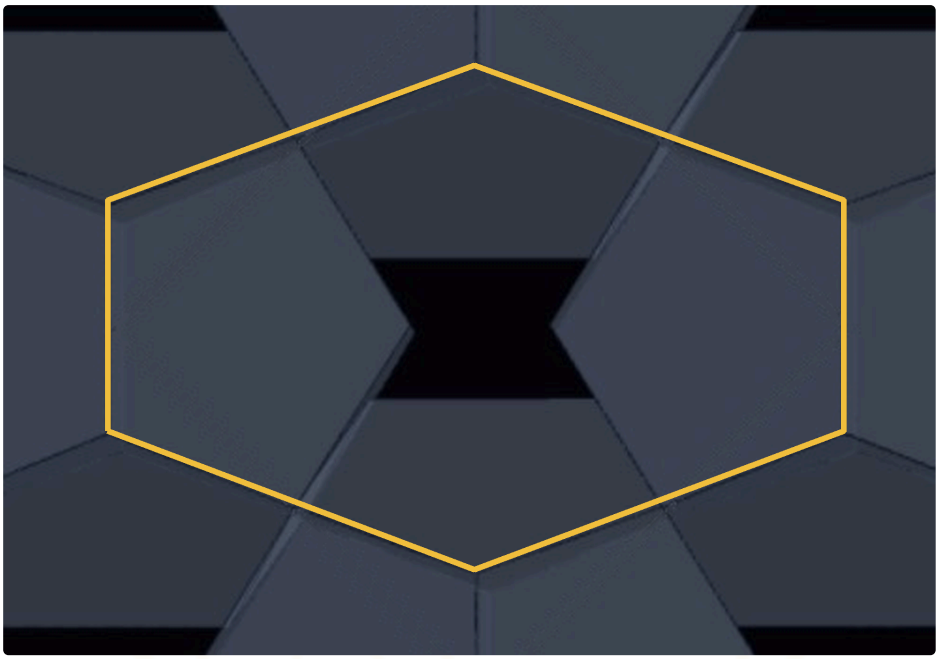


Figure 2. Where it started: DICTO's view into the dodecahedron layer of #47 in Kaleidohedra, with the long hexagon he saw outlined in gold. The void (black) lies at its centre, with two dodecahedra overlapping it on either side. (The app's cube outlines have been removed from the screenshot.)

2. The main result

> *Take any solid whose edges are all the same length, such as a dodecahedron. However you slide it, as long as you do not slide it along one of its own edges, every edge of the shape it sweeps comes out one edge-length long.*

> **The Polyhedral Stretching Theorem.** Let P be a convex polyhedron whose edges all have length 1, and slide it a distance 1 along a direction u. The swept shape (the convex hull of P and P moved by u) has every edge of length 1 unless u is parallel to an edge of P. Its faces are: each face of P not parallel to u, kept whole (one copy, at the end it faces); each n-sided face parallel to u, stretched into an (n+2)-gon, with one side fewer for each of its edges parallel to u; and one parallelogram for every edge of P's outline seen along u, a rhombus when the slide is 1.

> *What happens to the faces: the faces at the two ends stay just as they were; a face lying along the slide gets longer and gains two sides; and around the middle grows a belt of*

> *diamond shapes, one for each edge on the solid's outline.*

For the regular dodecahedron D, whose pentagons have no parallel edges, a pentagon lying along the slide becomes a hexagon if u runs along one of its edges and a heptagon otherwise, and, when no pentagon lies along the slide, the belt has 10 or 12 rhombi. The proof is short and is given in Appendix A. In 2 000 random directions the belt always had 10 or 12 rhombi, and every edge was 1 (Appendix B); the theorem proves the equal edges, while the count of 10 or 12 is what the sample showed.

3. Special directions

> *Some directions are special: they line up with the dodecahedron's own symmetry, and give the cleanest shapes.*

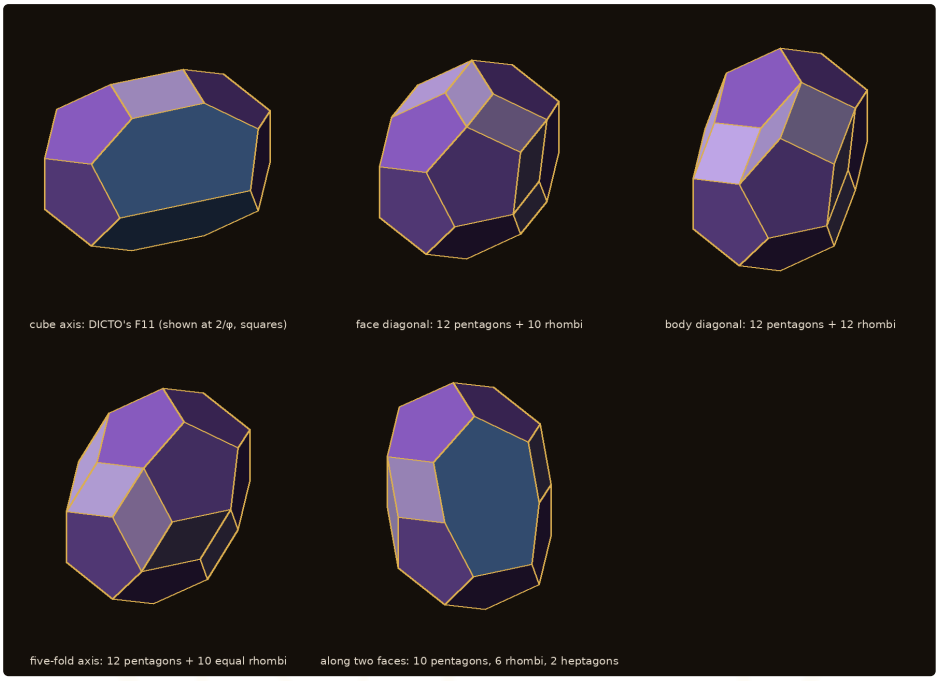


Figure 3. The five directions of the table. Pentagons violet, rhombi and rectangles lilac, hexagons and heptagons blue. The five-fold stretch is the most regular: its ten rhombi are all alike.

Slide along	Faces	Every edge 1?
a cube axis (DICTO's F11)	8 pentagons, 4 hexagons, 2 rectangles (squares at slide 1)	no: the slide runs along two edges
a face diagonal of Euclid's cube	12 pentagons, 10 rhombi	yes
a body diagonal (three-fold axis)	12 pentagons, 12 rhombi (six of them squares)	yes
a five-fold axis (through two opposite pentagons)	12 pentagons, 10 rhombi, all ten alike (angles 58° and 122°)	yes
a direction lying along two pentagons (example)	10 pentagons, 6 rhombi, 2 heptagons	yes

4. Equal-edged and telescoped

> *DICTO names the whole family the DICTO stretched solids: equal-edged at a stretch of one edge, telescoped at any other length. The faces stay the same; only the belt grows.*

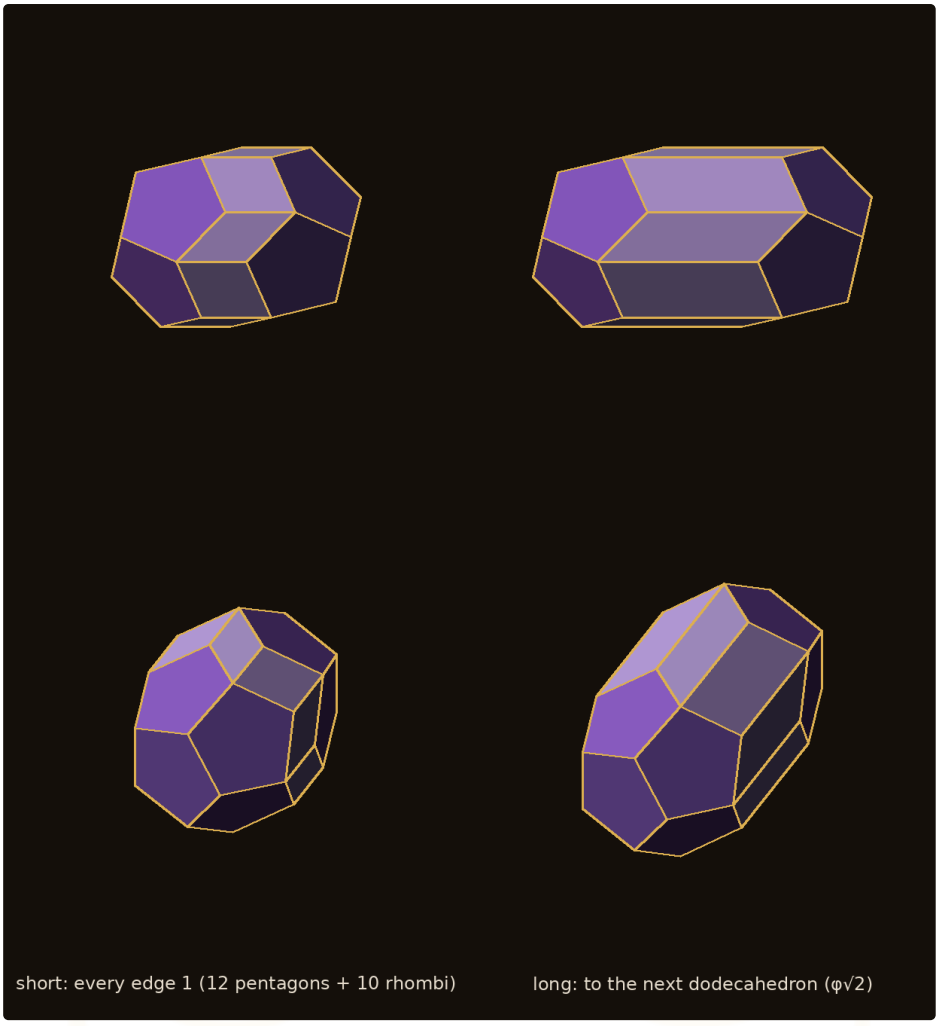


Figure 4. The face-diagonal slide. Left: equal-edged (slide 1), every edge 1, the rhombi of three shapes (2 with angles 45° and 135°, 4 with 55° and 125°, 4 with 69° and 111°). Right: telescoped to $\varphi\sqrt{2} \approx 2.288$, all the way to the next dodecahedron.

For any slide length the faces keep their kinds; only the belt's long sides change. That is why the telescoping model of §7 works: it is one solid at every length.

5. Three caps and four caps

> *Instead of sliding along a line, put a dodecahedron at each corner of a triangle or a tetrahedron and wrap them together. Equilateral triangles appear.*

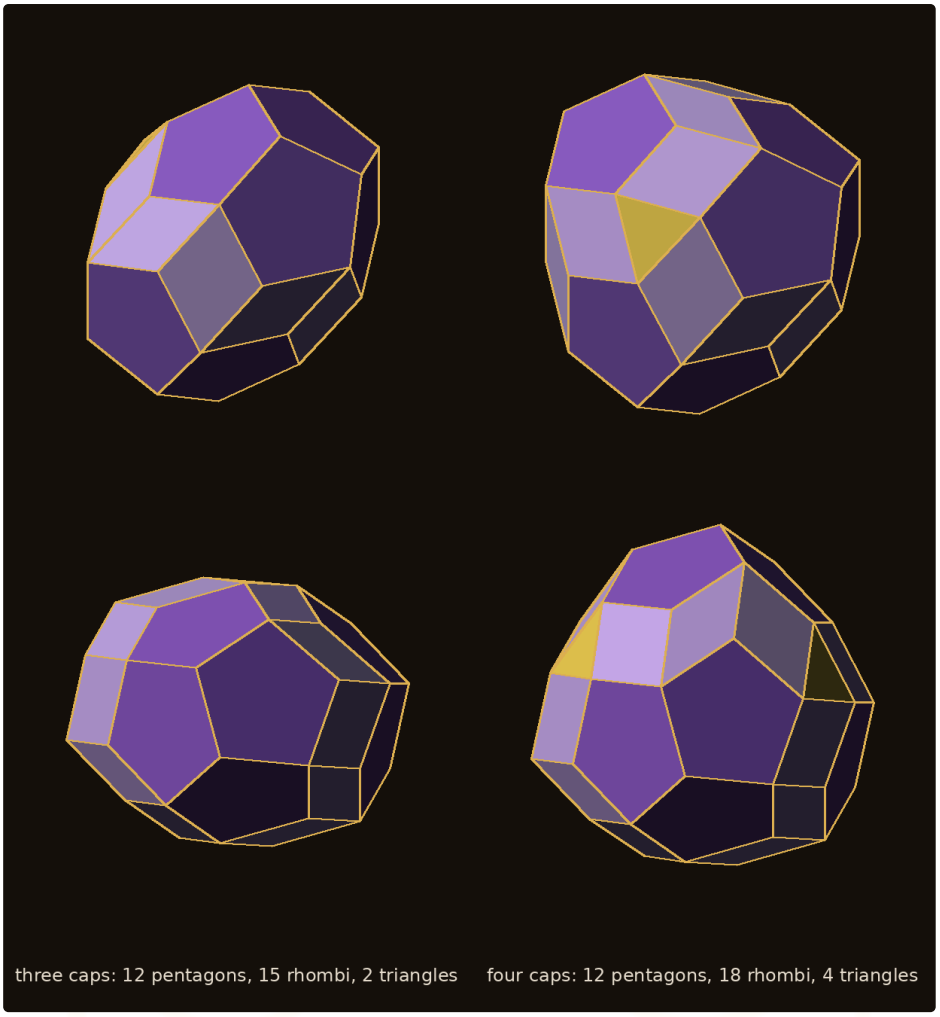


Figure 5. Dodecahedra at the corners of an equilateral triangle (left) and a regular tetrahedron (right), one edge apart; triangles gold.

Corollary. Wrapping dodecahedra placed at the corners of an equilateral triangle or a regular tetrahedron of side 1, in the directions of neighbouring cubes of Euclid's lattice, at distance 1, gives solids with every edge 1: three caps, 12 pentagons, 15 rhombi and 2 equilateral triangles (29 faces); four caps, 12 pentagons, 18 rhombi and 4 equilateral triangles (34 faces). (The same argument as the theorem: every edge is an edge of the dodecahedron or of the triangle or tetrahedron.)

DICTO also speculated that turning one end of the two-cap shape might make its belt into triangles, as in an antiprism. It cannot: over every turn and shift, the best band of triangles still has edges a third of an edge too long or short, because the rim zigzags. Three and four caps give the triangles instead.

6. The other Platonic solids, and the icosahedron family

> *The theorem works for every solid whose edges are all equal. Stretch the other Platonic solids and some famous shapes appear, and some new-looking ones.*

The Polyhedral Stretching Theorem needs only that every edge of the solid has the same length, so it applies to the tetrahedron, cube, octahedron and icosahedron as well (checked for 1 000 random stretches of each). Several results are well-known solids, which is a good sign that the stretch is a natural operation: the tetrahedron with four caps, at the corners of a tetrahedron turned the other way, is the cuboctahedron and the octahedron with four caps

the truncated tetrahedron (both Archimedean); the tetrahedron stretched through a corner is the elongated triangular pyramid, with three caps on a triangle turned against its base the triangular cupola, and the octahedron stretched through a corner the elongated square bipyramid (Johnson solids J7, J3 and J15); the stretched cube is a box, a hexagonal prism or a zonohedron.

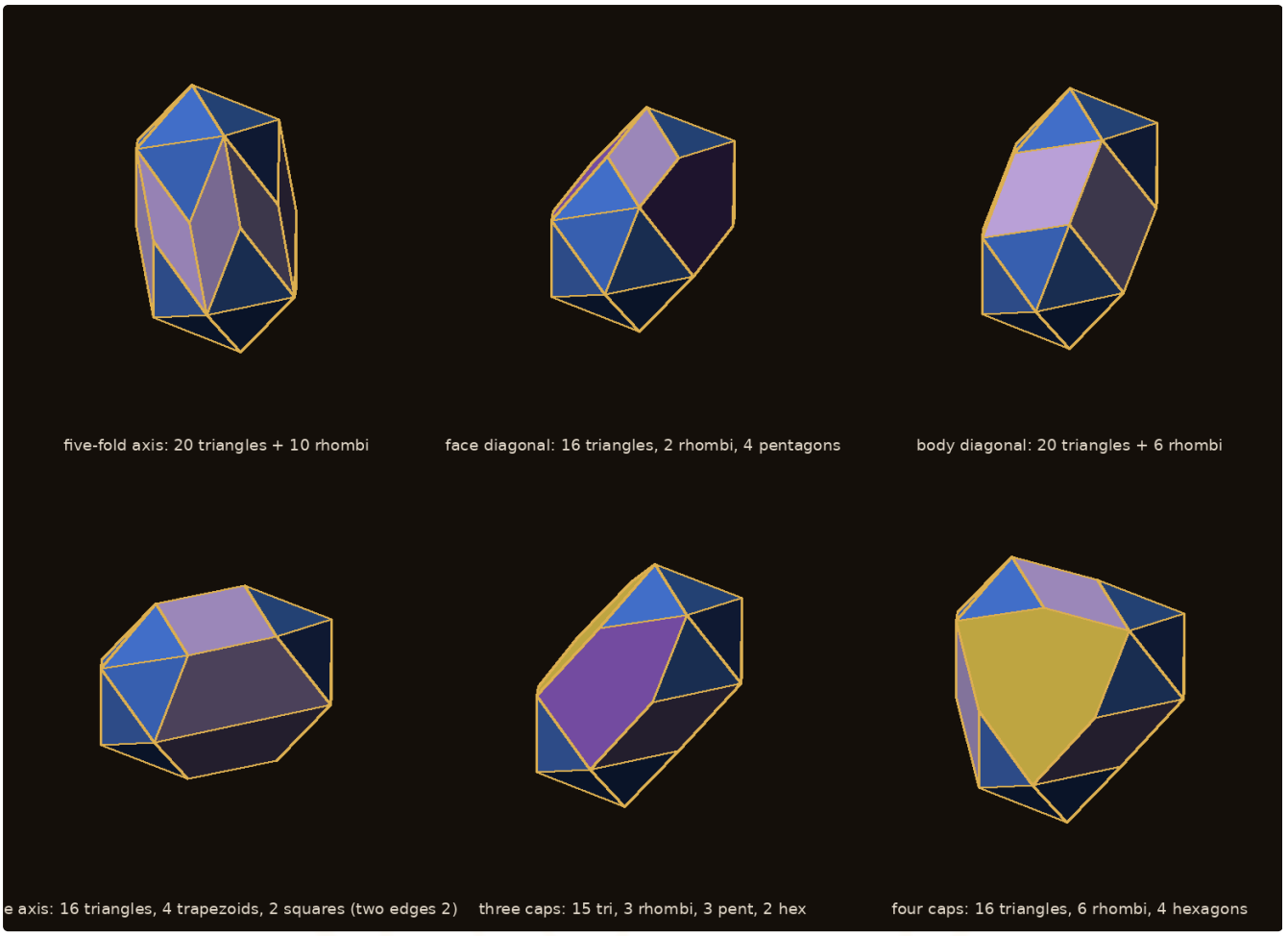


Figure 6. The icosahedron (edge 1) stretched one edge-length along four directions, and with three and four caps; every edge 1 except along the cube axis, which runs along two of its edges. Triangles blue, rhombi lilac, pentagons violet, hexagons gold.

Icosahedron	Faces
stretched along a five-fold axis	20 triangles, 10 rhombi
stretched along a face diagonal	16 triangles, 2 rhombi, 4 pentagons (the four triangles lying along the slide, each stretched into a pentagon)
stretched along a body diagonal	20 triangles, 6 rhombi
stretched along a cube axis	16 triangles, 4 trapezoids, 2 squares; not equal-edged: the axis runs along two edges, which become 2 long
three caps	15 triangles, 3 rhombi, 3 pentagons, 2 hexagons
four caps	16 triangles, 6 rhombi, 4 hexagons

The icosahedron family is the richest. None of its five equal-edged shapes is a Johnson or Archimedean solid (they have rhombi), and none was found in the sources checked; the nearest relatives are Hart and Picciotto's Minkowski sum of the compound of five tetrahedra, with 20 triangles and 60 rhombi [6], and Ed Baumann's equal-edged near-misses, which also merge triangles into rhombi [8]. The octahedron stretched through a face (8 triangles, 6

rhombi; the same face counts as Hart and Picciotto's sum of two tetrahedra [6], though whether the shapes match was not checked) and the tetrahedron and cube in their other directions are also not among the named solids we checked; a short table of the Archimedean solids is left for later.

7. Make your own

> *Print, cut, fold and glue: every solid here can be held in the hand.*

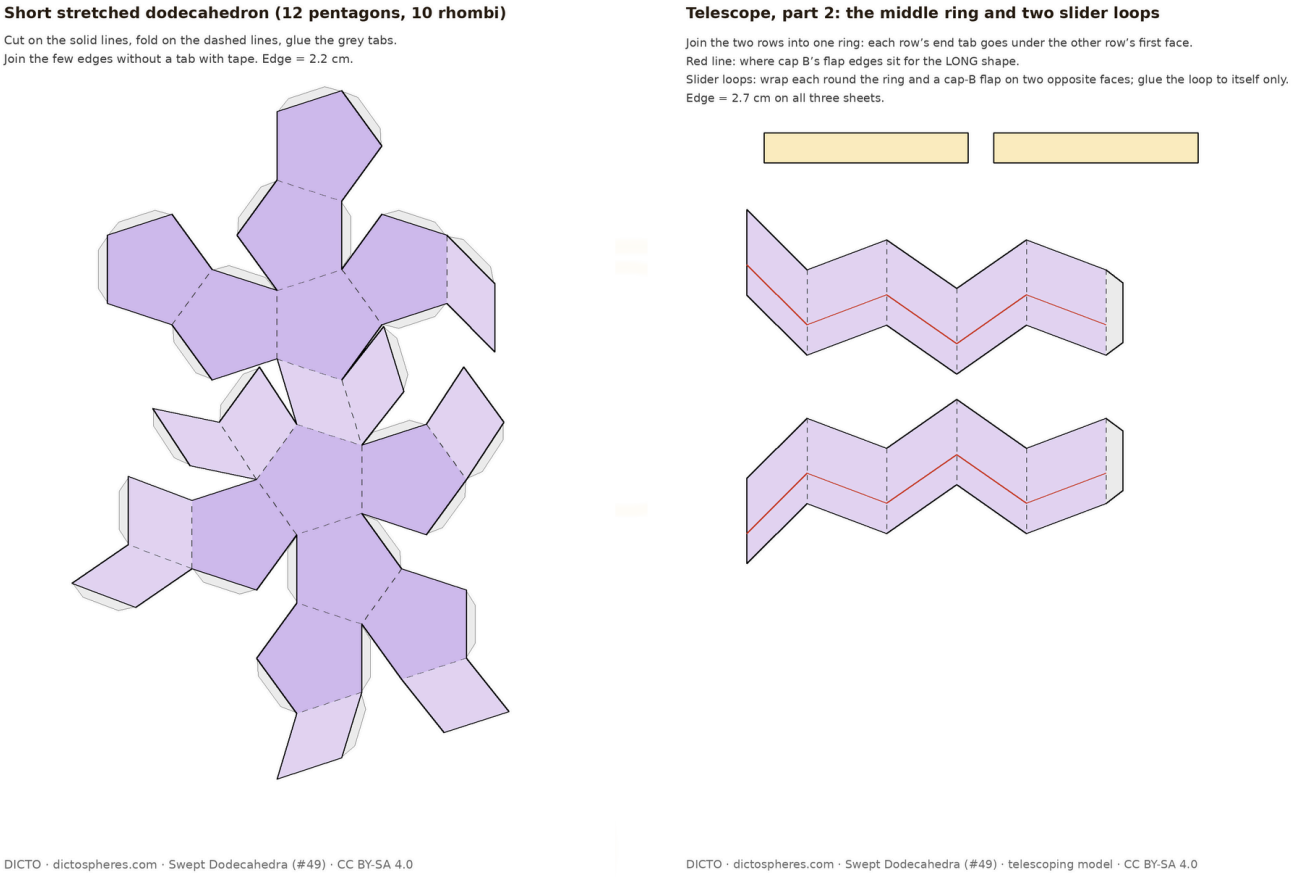


Figure 7. Left: the net of the short stretched dodecahedron. Right: the middle ring of the telescoping model.

Two files are attached. *swept-dodecahedra-nets.pdf*: five A4 nets, the short and long stretched dodecahedra, the three- and four-cap forms and DICTO's F11 (square version), with fold lines dashed, cut lines solid and glue tabs; the few edges without a tab are joined with tape. *swept-dodecahedron-telescope.pdf*: DICTO's telescoping model in three parts, a cap with inner flaps, a loose middle ring of ten belt faces, and a cap with outer flaps. The faces slide over and under each other; two slider loops keep them in line. Pushed together it is the short shape; pulled to the red marks it is the long one; it extends to about 2.7 edges.

8. What is new, and what is known

> *The method is old; these shapes were not found named anywhere we looked.*

Sliding a solid along a line, or adding two solids corner by corner, is a standard construction (the Minkowski sum); how a face behaves when the slide lies in its plane, and how a belt forms around the outline, are known in general [4, 5]. Hart and Picciotto's sums of tetrahedra build triangle-and-rhombus solids the same way, without pentagons [6]; the convex solids glued from regular pentagons alone are listed in [7]. In the sources checked (the reference

library and web searches on 22 October 2026) we did not find the dodecahedron swept along a face diagonal, the body diagonal or a five-fold axis (the last, with its ten equal rhombi, deserves a further search), nor the three- and four-cap forms, named or described. Since the construction is simple, they may exist somewhere unnamed; we present them as DICTO's family on that understanding.

9. Conclusion

> *One idea, sliding a dodecahedron, gives a whole family of solids with equal edges, and you can build them.*

The theorem shows why the DICTO stretched solids keep their edges equal: every edge is either an old edge or a copy of the slide. The face-diagonal slide keeps all twelve pentagons, and three and four caps add equilateral triangles. The same theorem holds for every solid with equal edges: among the Platonic solids it rediscovers the cuboctahedron, the truncated tetrahedron and three Johnson solids, and gives a family of new-looking stretched icosahedra.

> *Two things are still to find out: what happens when the same stretch is tried on the thirteen Archimedean solids, and whether the dodecahedron with ten equal diamonds has been seen before.*

Open questions. (1) The Archimedean solids also have equal edges, so the theorem applies to them; which of their stretches and caps are known solids, and which are new, is left for a later paper. (2) The dodecahedron stretched along a five-fold axis, with its ten congruent rhombi, deserves a wider prior-art search than the one made here.

Appendix A. Proof of the Polyhedral Stretching Theorem

> *Why it works: every edge of the stretched shape is either one of the solid's own edges, moved, or a copy of the slide itself. Both are one edge-length long, so every edge is too, unless an old edge and the slide point the same way and join into one double-length edge.*

The swept shape is the Minkowski sum $P \oplus S$ of P and the segment S from 0 to u. Every face of a Minkowski sum is the sum of a face of each summand, taken in a common outward direction n [4]. For S, the face in direction n is its end u if $n \cdot u > 0$, its start 0 if $n \cdot u < 0$, and S itself if $n \cdot u = 0$. Hence: a face f of P with normal n, $n \cdot u \neq 0$, appears as a translate of f; a face with $n \cdot u = 0$ appears as $f \oplus S$, the n-gon f widened by u, which gains the two copies of S at its extreme corners across u, and so has $n + 2$ sides, less one for each edge of f parallel to u (such an edge is lengthened rather than joined by a new side); an edge e of P on the outline seen along u (between a face with $n \cdot u > 0$ and one with $n \cdot u < 0$) appears as the parallelogram $e \oplus S$. The edges of $P \oplus S$ are therefore translates of edges of P (length 1) and translates of S (length $|u| = 1$), except where an edge of P is parallel to u: there $e \oplus S$ is a single segment of length 2. So every edge has length 1 exactly when no edge of P is parallel to u. ∎ The corollary is the same argument with S replaced by the triangle or tetrahedron, whose edges have length 1 and are not parallel to edges of the dodecahedron in the positions used.

Appendix B. Checks

> *Besides the proof, the computer stretched each solid in thousands of random directions and measured every edge; the paper nets were checked to fold flat without overlaps.*

In studies/2026-10-22-swept-dodecahedra: void_volume.py (the void between two dodecahedra on edge-sharing cubes: volume $\varphi^3 + \frac{1}{2}$, no overlap); sweep_theorem.py (2 000 random directions, all equilateral; belts of 10 or 12 rhombi; the four special directions); caps34.py (three and four caps: face kinds, edge lengths); gyro_caps.py (turned caps, for the face-diagonal and the true five-fold rims: no triangle belt, worst edge off by 0.336 at best); ring1.py and identify.py (the other four Platonic solids: 1 000 random stretches each, special directions, caps, and which results are known solids); icosa_family.py (Figure 6); nets.py and telescope.py (the printable nets: unfoldings checked to lie flat with no faces overlapping). Python 3 with numpy and scipy.

References

1. DICTO, Every Mix of the Platonic Solids Fills Space, #47, DICTO DOI: 10.5281/zenodo.23551326.
2. Euclid, The Thirteen Books of the Elements, tr. T. L. Heath, vol. III, Cambridge (1908), Book XIII, Proposition 17.
3. DICTO, Kaleidohedra by DICTO, finding F11 "The stretched dodecahedron" (2026-10-08), DICTO DOI: 10.5281/zenodo.23173809.
4. Diameter, decomposability, and Minkowski sums of polytopes, arXiv:1806.07643 (faces of a Minkowski sum as sums of faces).
5. Minkowski sum of a Voronoi parallelotope and a segment, arXiv:1501.01212 (how faces change when a segment is added).
6. G. W. Hart and H. Picciotto, Green Giants, in Zome Geometry, https://www.georgehart.com/zomebook/green-giant.html.
7. A complete list of all convex polyhedra made by gluing regular pentagons, arXiv:2007.01753.
8. Ed Baumann, Modified Johnson Near Misses, https://superliminal.com/geometry/near_misses/index.html.